

How to Select a 3D Printer Under ₹ 100,000



Abhishek A. Mutha is a senior technical correspondent at EFM

The biggest advantage of 3D printers today is rapid prototyping. At times, 3D printers could be the right equipment for low-batch production than going for traditional manufacturing. "3D printers started being used as geek tools and are now shaping up as user-friendly tools; even my eight-year-old daughter uses one for her school projects," says Vishal J.C., CEO, Grey Sim Ltd.

Shiv Bhambri, CEO, RS Components & Controls Ltd, feels, "3D printers are ideal for anyone who wishes to explore the fascinating world of 3D printing, build

prototypes or even carry out small volume production."

In this article, we particularly focus on 3D printers under ₹ 100,000 and attempt to help you make the right choice.

Generalising the family of 3D printers, Raja Sekhar Upputuri, co-founder, Think3D, notes that there are three kinds of 3D printers. "First, basic printers coming at an entry price point with very low build volume and a single extruder; second, medium range printers with decent build volume and single or dual extruders; and third, printers with high build volume used for semi-industrial and semi-professional applications, which, of course, come with a higher price tag."

Upputuri says, "Companies that started with basic printers are venturing into the other two categories, and vice versa. Customers have a variety of options to choose from today. In fact, some vendors are introducing stripped-down, economical versions of 3D printers available for as low as ₹ 30,000." Although these printers might not provide professional-grade experience, these are mostly used for educational purposes by schools and colleges.

What is new in 3D printers under ₹ 100,000

The only technology being offered for 3D printers under ₹ 100,000 is fused deposition modelling (FDM). "The major advancement happening with respect to FDM is scaling down of the cost of 3D printers using this technology," informs Prudhvi Reddy, co-founder, Think3D.

"For printers within or even slightly exceeding the price tag of ₹ 100,000, there are plenty of options available to choose from. This is mostly because of the stabilisation of FDM technology, but there is no major breakthrough in the printers as of now," Reddy adds.

TABLE I

Select The Right 3D Printer Based on Your Needs

Build a 3D printer (RepRap printers)	Prototyping 3D printers	Low-cost industrial printers
Prusa Mendel (₹ 44,400*)	My3DBot (₹ 90,000)	There is no low-cost industrial printer for under ₹ 100,000;
Prusa i3 complete kit (₹ 47,700*)	Brahma3 Anvil (₹ 100,000)	an industrial 3D printer costs anywhere
Prusa 20.3cm (8") i3 kit (₹ 54,100*)	Gbot 3D Creator (₹ 66,666*)	from ₹ 300,000 to ₹ 3 million
	IdeaWerk (₹ 75,000)	

*Price excluding taxes and shipping charges

Other exciting advancements and technologies in 3D printers

There are some exciting changes in printers costing between ₹ 200,000 and ₹ 300,000.

Multi-colour printing. Earlier, only single- or double-colour objects could be printed. Today, multiple-coloured objects can be printed using red-green-blue (RGB) colour combinations. This is one major advancement in 3D printers using FDM technology. Multi-colour is the need of the hour.

New printing materials. Printing material used to be mainly plastic or nylon but today composite materials like carbon fibre are entering the market. Carbon fibre is a very strong material that is frequently used in the automotive sector. Printers using such materials are priced at around ₹ 300,000.

Exciting technologies. Apart from FDM, 3D printers use two other popular technologies called stereolithography (SLA) and selective laser sintering (SLS). Prices of printers using SLA are coming down drastically and are slowly entering the consumer domain. SLA gives a better finish and accuracy when compared with FDM.

SLS uses metal to create solid objects, although the cost of printers using SLS is still on the higher side. It is purely used for industrial applications in segments such as medical, automotive, aerospace and defence, to name a few. For instance, in medical, SLS 3D printers are being used to create and replace portions of a damaged skull. Mahindra Reva is already using this technology to print automobile components. Rolls Royce is using it for their aircraft engines.

—Raja Sekhar Upputuri, co-founder, Think3D